CLAIMS

What is claimed is:

1. A method for encoding data on a parallel communication interface having a plurality of physical channels, the method comprising the steps of:

(a) defining a logical channel as a group of 'n' physical channels from the plurality of physical channels, where n is greater than 1;

(b) determining a combined state-space of the logical channel based on possible states of the 'n' constituent physical channels; and

(c) encoding a single data block by mapping it to a single unique state within the combined state-space of the logical channel.

2. The method of claim 1, wherein the physical channels are pixels in a two-dimensional array.

3. The method of claim 2, wherein $n = 2$, defining a "bigram" logical channel.

4. The method of claim 3, wherein each pixel is capable of representing 2^{24} states, and the bigram logical channel is capable of representing 2^{48} states.

5. The method of claim 4, wherein the single data block is a 6-byte data block.

6. The method of claim 1, wherein the parallel communication interface is a massively parallel optical interface as described in U.S. Patent Application No. 19/559,705.